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Trustworthy uncertainty quantification for quantitative and diffusion body MRI, and its decision-use in MR-guided adaptive radiotherapy: a field review

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Abstract

Quantitative and diffusion body MRI increasingly delivers not an image but a *number* — a tissue parameter offered to a clinical decision — and, in MR-guided adaptive radiotherapy, a number that can steer dose. This review asks whether the *uncertainty* attached to that number is trustworthy enough to act on. We organise the field along the spine a clinician implicitly walks: *trust* (is the error bar honest?), *value of information* (does a trusted bar change the decision?), and *action* (what happens when it is deployed to drive dose?), over a foundations layer (what is measured, and how) and beneath a gap map. Synthesising the literature across estimation, calibration, decision analysis, and MR-guided radiotherapy, we find that trust is conditional: broadly defensible for the diffusion coefficient D and ADC, but fragile for the perfusion parameters f and D^* , where measured test–retest reliability is poor and where deep-learning uncertainty quantification typically reports only marginal, aggregate calibration. The decision-analytic apparatus to price a calibrated interval — net benefit, decision calibration, value of information — is mature but rarely connected to quantitative-MRI uncertainty; and the propagation of *biomarker* uncertainty into the adaptive dose decision remains the immature link. We close with four open problems sited at the seams between trust, value, and action. This review reports no new measurement; its contribution is synthesis, taxonomy, and gap identification.

1 Scope and framing

Quantitative and diffusion body MRI — intravoxel incoherent motion (IVIM), diffusion-weighted ADC/DKI, dynamic contrast-enhanced (DCE) perfusion, and relaxometry — increasingly produces not just an image but a *number*: a tissue parameter offered to a clinical decision. The question this review surveys is whether the **uncertainty** attached to that number is trustworthy enough to act on, and what happens when it is used to steer **MR-guided adaptive radiotherapy** (MR-Linac / MRgART).

We organise the literature along the spine a clinician implicitly walks:

1. **Trust** — is the error bar honest (calibrated, covering)?
2. **Value of information** — given a trusted bar, does it change the decision?
3. **Action** — when deployed to drive dose, where does it hold and where does it break?

with a **foundations** layer beneath (what is measured, and how) and a **gap map** above. The survey covers UQ for quantitative/diffusion body MRI and its decision-use in adaptive RT; it excludes non-quantitative MRI, neuro-only work, generic computer-vision UQ, and non-MR-guided RT (the scope boundary is recorded in the repository’s `ASSUMPTIONS.md`). Table 1 lays out the survey axis and the thematic buckets it induces.

Table 1. The survey axis. The field is classified by the question each work answers about *somebody's* uncertainty estimate; a cross-cutting gap map (Sec. 6) sits over the three axes at the seams between them.

axis	what the bucket asks
Foundations	What is the estimand, and what estimator (with what uncertainty) produced it? (Sec. 2)
Trust	Is the error bar honest — calibrated and covering? (Sec. 3)
Value of information	Given a trusted bar, does it change an oncologic decision? (Sec. 4)
Action	When deployed to drive dose on an MR-linac, where does it hold and break? (Sec. 5)
Gap map	Where, at the seams between the axes, do the field's UQ-trust questions remain open? (Sec. 6)

2 Foundations: estimands and estimators

Before honesty can be assessed, the estimand and estimator must be fixed. The IVIM model itself was introduced to separate molecular diffusion from capillary-network microcirculation within a voxel [1, 2], treating randomly oriented capillary flow as a pseudo-diffusion process [3]; DCE-MRI quantification rests on the compartmental transfer-constant model [4] and its standardised quantities K^{trans} , v_e , k_{ep} [5].

The estimator is not innocent. Cramér–Rao/Fisher analysis shows the achievable precision of IVIM D and f is strongly SNR-dependent [6], and the same bounds can be inverted to design SNR-efficient acquisitions [7]. Empirically, IVIM parameter estimates differ *significantly* across fitting algorithms in the upper abdomen, with Bayesian fitting giving the lowest inter-reader and inter-subject variability [8]; simulation [9] and pancreatic-cancer data [10] corroborate the Bayesian advantage for D and f . A parallel line replaces hand-built fits with learned ones — neural-network IVIM/kurtosis mapping [11], supervised and unsupervised physics-informed deep fitting [12, 13], and self-supervised forward-model estimation [14] — with recent work decomposing predictive uncertainty into aleatoric and epistemic components [15]. DCE shares the trajectory, e.g. Bayesian spatial tracer-kinetic estimation [16]. **Takeaway:** the uncertainty a method reports is inseparable from the estimator that produced it, so trust must be assessed per-estimator, not per-biomarker.

3 Trust: is the error bar honest?

3.1 The general machinery of trust

The UQ community offers a toolkit for honest uncertainty that the imaging literature draws on. Distribution-free **conformal prediction** wraps any black-box predictor in a set/interval valid at a user-chosen error rate [17, 18]. **Calibration** methods recalibrate regression intervals post-hoc [19] and correct the systematic over-confidence of modern deep networks [20]. Practical deep UQ families — deep ensembles [21], MC-dropout as approximate Bayesian inference [22], and single-pass evidential regression [23] — give usable uncertainty, but their quality **degrades under dataset shift**, so trust must be tested against shift, not only in-distribution [24].

3.2 The metrology of trust in quantitative imaging

Imaging has its own standards for honest reliability. The QIBA framework anchors technical performance on three metrology areas — linearity/bias, repeatability, reproducibility [25] — with companion guidance on comparing algorithms [26] and on consistent terminology [27]. Reliability reporting leans on the intraclass correlation coefficient [28] and on limits-of-agreement rather than correlation for method comparison [29].

3.3 Where trust is empirically tested in body MRI

When these standards are applied to body diffusion/perfusion MRI, the error bars are often **not** honest — and the dishonesty is parameter- and cohort-dependent. ADC is a comparatively reliable biomarker but its in-vivo reproducibility is markedly worse than phantom reproducibility [30], and the body-DWI consensus already urged baseline reproducibility studies as part of study design [31, 32]. The perfusion parameters are the weak point: short-term test–retest repeatability coefficients for IVIM f and D^* reach $\sim 126\%$ and $\sim 197\%$ in rectal cancer [33], D^* reproducibility is only moderate (ICC ≈ 0.55 , CV $\approx 20\%$) [34], and multi-scanner liver IVIM within-subject CVs run from $\sim 5\%$ for D up to $\sim 14\%$ for perfusion-fraction components [35]. Cross-modal anchoring does not rescue D^* : its correlation with DCE K^{trans} is weak in one rectal cohort ($r = 0.389$) [36] and non-significant in another [34]. **Takeaway:** trust in body-MRI UQ is conditional — broadly

defensible for D/ADC , fragile for f/D^* — yet most deep-learning UQ reports aggregate or marginal calibration [15], leaving the conditional failure region under-characterised.

4 Value of information: does a trusted bar change the decision?

A calibrated interval earns its cost only if it moves an action. The decision-analytic literature supplies the apparatus: **decision curve analysis / net benefit** evaluates a model by clinical net benefit across the decision threshold [37–39], where the threshold encodes the clinician’s relative weighting of a missed disease against an unnecessary intervention; performance frameworks argue such decision-analytic measures should be reported whenever a model supports decisions [40], and relative-utility curves quantify how much achievable utility a predictor captures [41]. From the inference side, **decision/loss-calibrated** methods tie the posterior approximation to the downstream decision — decision calibration [42] and loss-calibrated Bayesian inference [43, 44]. From health economics, **value-of-information** analysis (EVPI) prices a measurement by combining the probability it changes a decision with the benefit of that change [45], under the decision-theoretic stance that choices follow expected net benefit rather than statistical significance [46]. **Takeaway:** the machinery to ask “does this body-MRI interval change an oncologic decision?” exists and is mature — but is rarely connected to quantitative-MRI uncertainty.

5 Action: deployment in MR-guided adaptive radiotherapy

The action setting where a quantitative-MRI biomarker most directly drives dose is the MR-Linac. Integrating diagnostic-quality MRI with a linear accelerator [47, 48] matured into a clinically feasible 1.5 T MRI-guided system [49, 50] and an emerging paradigm for adaptive radiation oncology [51, 52]. The biological premise is **dose painting** — prescribing a non-uniform dose from a biological target volume derived from functional/molecular images [53–55] — with DWI/ADC a leading candidate response biomarker [56] and quantitative MRI proposed to update the plan to observed tumour response [51, 57]. Two cautions recur. First, **uncertainty must be carried into the optimisation:** robust/probabilistic planning incorporates motion and uncertainty directly rather than relying on PTV margins [58]. Second, **technical validation gates clinical use:** imaging biomarkers on an MR-linac require repeatability/reproducibility validation before driving decisions, because the integrated scanner differs from a diagnostic system [59]. **Takeaway:** the field is converging on biomarker-driven adaptive dose, but the propagation of *biomarker uncertainty* into the dose decision is the immature link.

6 Gap map: open problems

The field’s unresolved UQ-trust questions cluster at the **seams between the axes** — which is also, stated as honest positioning rather than as this review’s organising centre, where the author’s own program sits.

- **G1 — conditional honesty (trust seam).** Repeatability evidence shows f/D^* are far less reliable than D/ADC [33–35], yet deep-learning IVIM-UQ typically reports marginal/aggregate calibration [15]. *Open:* calibration conditioned on the regime where it fails, and conformal/coverage guarantees that respect it [18, 24].
- **G2 — from calibration to decision (trust→value seam).** The decision-analytic toolkit [37, 42] is rarely applied to quantitative-MRI uncertainty. *Open:* when does a calibrated body-MRI interval actually change an oncologic decision versus merely being reported?
- **G3 — uncertainty into dose (value→action seam).** Adaptive-RT reviews call for technically validated biomarkers to steer dose [51, 59], and robust planning can ingest uncertainty [58]. *Open:* end-to-end propagation of *biomarker* uncertainty (not just geometric/motion uncertainty) into the adaptive dose decision, and the identifiability walls that cap it.
- **G4 — estimator-dependence of trust (cross-cutting).** Reported uncertainty depends on the estimator [8, 12]. *Open:* standards that make UQ comparable across classical and learned estimators, extending QIBA metrology [25, 27] to the uncertainty itself, not only the point estimate.

7 Honest scope and discussion

This review’s contribution is the **synthesis, taxonomy, and gap map** above — not new data. It asserts no experimental result; every empirical statement is attributed to a verified external source.

Where the author’s own program is relevant it is named only as positioning within G1–G4, never as the organising thread.

As a portfolio artifact the review is deliberately *trigger-independent*: it strengthens a first research application by demonstrating field command and supplying a citable synthesis, and it depends on none of the author’s own results publishing. In this role it **absorbs** what an earlier plan called a separate portfolio-thickening “Buttress” note: rather than a stand-alone companion, the field command and the open-problem map are folded here, and the gap map (Sec. 6) doubles as the positioning of the author’s program within the field’s open seams — honest positioning, not the survey’s organising centre. The four gaps are stated as the *field’s* open problems; the author’s contributions, where they exist, sit inside G1–G4 on equal footing with external work and are not asserted here as published results.

The dominant failure mode for a review of this kind is the phantom or mis-quoted citation. The repository makes that mechanical: every entry in the citation base carries a resolvable identifier (DOI, arXiv eprint, or stable proceedings URL) and a one-line verified claim, and a citation gate fails the build on any unresolved identifier, any orphan between the bibliography and the verified-claim ledger, or any `\cite` in this manuscript with no backing entry. Four reliability/oncology foundations are flagged at thesis level: their central claim is title/abstract-evident, but any load-bearing *number* from them is re-pulled verbatim from the source before use rather than approximated.

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Data availability

This review reports no new experimental data. The verified citation base (`limbo.bib`), the per-citation verified-claim ledger (`CITATIONS.md`), and the citation gate (`verify_citations.py`) that enforces resolvable identifiers and zero phantom citations are openly available in the project repository; one command (`./reproduce.sh`) re-runs the gate, the test-suite, and the manuscript build.

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